

Essential Fell Ideals and the Ball-Covering Property

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

A family $\mathcal{U}$ of nonempty open subsets of a topological space X is π -dense if every nonempty open subset of X contains a member of $\mathcal{U}$. Equivalently, $\mathcal{U}$ is a π -basis.

For an open set $O \subset \text{Prim}(A)$, let A_O denote the corresponding closed two-sided ideal of A .

Statement

Theorem 1 (π -dense atomic chart criterion). *Let A be a C^* -algebra. Suppose there is a countable π -dense family (O_n) of open subsets of $\text{Prim}(A)$ such that*

$$A_{O_n} \cong C_0(O_n, \mathcal{K}(H_n))$$

for separable nonzero Hilbert spaces H_n . Then A has UBCP.

Corollary 1 (essential Fell ideal criterion). *Let A be a C^* -algebra. Suppose A has an essential ideal J which is Fell, and suppose every irreducible representation of J acts on a separable Hilbert space. Then*

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(A) \text{ has a countable } \pi\text{-basis}.$$

Idea of the Proof

The atomic multiplier patch proof does not need good charts everywhere. For each $a \in S_A$, the set

$$W_a = \{P \in \text{Prim}(A) : \|a + P\| > 3/4\}$$

is nonempty and open. A countable π -dense family of atomic charts gives one chart $O_n \subset W_a$, and the rank-one coefficient construction from the Fell packet covers a using a centre supported in that chart.

If J is essential, then $\text{Prim}(J)$ is a dense open subset of $\text{Prim}(A)$. Therefore any countable π -basis of $\text{Prim}(A)$ can be refined inside $\text{Prim}(J)$. Since J is Fell with separable irreducible representations, its spectrum is locally Hausdorff and its Hausdorff patches are continuous trace with separable fibres. Local triviality then gives the required π -dense atomic charts.

Proof

Lemma 1 (rank-one coefficient estimate). *Fix $0 < \theta < 1$. There are constants $M > 1$ and $\rho < M$ such that, whenever $T \in B(H)$ is a contraction, $\xi, \eta \in H$ are unit vectors, and*

$$\text{Re}\langle T\xi, \eta \rangle \geq \theta,$$

then, for $Ru = \langle u, \xi \rangle \eta$,

$$\sup_{0 \leq t \leq M} \|T - tR\| \leq \rho.$$

Proof. This is Lemma 1 of the atomic multiplier chart packet [5]. The proof is a two-region estimate on a unit vector $x = \lambda\xi + v$, $v \perp \xi$. If $|\lambda|$ is bounded away from one, the large rank-one term is already smaller than t by a fixed factor; if $|\lambda|$ is close to one, the positive coefficient $\text{Re}\langle T\xi, \eta \rangle$ subtracts a fixed linear amount from the square norm. For large M this gives a uniform $\rho < M$. $\square$

Proof of Theorem 1. Fix $\theta = 1/2$, choose M, ρ from Lemma 1, and put

$$r = \frac{M + \rho}{2}.$$

Thus $\rho < r < M$.

For each n , fix an isomorphism

$$\Phi_n : A_{O_n} \rightarrow C_0(O_n, \mathcal{K}(H_n)),$$

a countable dense set D_n in the unit sphere of H_n , and a countable family $\mathcal{B}_n \subset C_c(O_n)$, $0 \leq h \leq 1$, $\|h\| = 1$, whose supports are cofinal in nonempty open subsets of O_n . These bump families exist because the O_n 's are locally compact Hausdorff and the countable π -dense family restricts to a countable π -basis on each O_n .

For each n , $h \in \mathcal{B}_n$, and $\xi, \eta \in D_n$, define

$$c(n, h, \xi, \eta) = \Phi_n^{-1}(Mh(\eta \otimes \xi^*)) \in A_{O_n} \subset A.$$

These centres are countable and have norm M .

Let $a \in S_A$, and set

$$W_a = \{P \in \text{Prim}(A) : \|a + P\| > 3/4\}.$$

This is nonempty open, so choose $O_n \subset W_a$. Pick $P_0 \in O_n$. In the multiplier representation of a over

$$A_{O_n} \cong C_0(O_n, \mathcal{K}(H_n)),$$

let $T_P \in B(H_n)$ denote the fibre multiplier. Then $\|T_{P_0}\| > 3/4$, so choose unit vectors ξ_0, η_0 with

$$\text{Re}\langle T_{P_0}\xi_0, \eta_0 \rangle > 2/3.$$

Approximate them by $\xi, \eta \in D_n$ so that the coefficient remains larger than $1/2$.

Multiplier fields are strict-continuous, hence fixed scalar matrix coefficients are continuous. Therefore

$$\text{Re}\langle T_P\xi, \eta \rangle \geq 1/2$$

on some nonempty open $N \subset O_n$. Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$, and put $c = c(n, h, \xi, \eta)$.

For $P \in \text{supp } h$, Lemma 1, applied with $t = Mh(P)$, gives

$$\|a + P - (c + P)\| \leq \rho < r.$$

For $P \notin \text{supp } h$, the centre vanishes in the primitive quotient at P , so

$$\|a + P - (c + P)\| = \|a + P\| \leq 1 < r,$$

after increasing M , if necessary. Taking the supremum over primitive quotients gives $\|a - c\| < r$. The countably many balls

$$B(c(n, h, \xi, \eta), r)$$

therefore cover S_A , and since every centre has norm $M > r$, A has UBCP. $\square$

Lemma 2 (essential ideals give dense primitive support). *For an ideal $J \triangleleft A$, J is essential if and only if $\text{Prim}(J)$, identified with the open subset*

$$\{P \in \text{Prim}(A) : P \not\supset J\},$$

is dense in $\text{Prim}(A)$.

Proof. The complement is the hull of J . If this hull had nonempty interior, the corresponding nonzero ideal would be disjoint from J , so J would not be essential. Conversely, if a nonzero ideal is disjoint from J , its primitive support gives a nonempty open subset of the hull of J . Thus essentiality is equivalent to the hull having empty interior, i.e. to $\text{Prim}(J)$ being dense. $\square$

Proof of Corollary 1. If A has BCP, the primitive π -base obstruction [6] gives a countable π -basis for $\text{Prim}(A)$. UBCP implies BCP, so it remains to prove UBCP from the countable π -basis assumption.

By Lemma 2, $\text{Prim}(J)$ is dense open in $\text{Prim}(A)$. Since J is Fell, $\text{Prim}(J)$ is locally Hausdorff; on Hausdorff open patches, Fell algebras have continuous trace [2]. Local triviality of continuous-trace algebras and separability of the irreducible representations give, inside every nonempty open subset of $\text{Prim}(J)$, a nonempty open patch O such that

$$J_O = A_O \cong C_0(O, \mathcal{K}(H))$$

for separable nonzero H .

Now refine a countable π -basis of $\text{Prim}(A)$ by these patches inside the dense open set $\text{Prim}(J)$. The resulting countable family is π -dense in $\text{Prim}(A)$ and satisfies the hypotheses of Theorem 1. Hence A has UBCP. $\square$

Scope and Limitations

This theorem is stronger than the Fell-algebra packet because A itself need not be Fell, type I, liminal, or separable. The quotient by the essential Fell ideal can be arbitrary. The theorem does not handle the case where the largest separable-Fell ideal is not essential; then the complement has nonempty primitive interior and must be analyzed as a genuine remaining piece.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, essential Fell ideals, π -dense Fell ideals, atomic multiplier charts, type-I beyond Fell, BCP, and UBCP. Web searches on June 17, 2026 for largest Fell ideals and Fell points found nearby ideal-structure literature including an Huef–Williams and the Clark–an Huef–Raeburn Fell reference [2], but no BCP theorem of this form. The result is recorded as an agent-produced partial theorem, subject to human literature review.

Human Review Focus

Reviewers should check:

1. the π -dense version of the atomic multiplier chart proof;
2. the essential ideal/dense primitive support equivalence;
3. the refinement from an essential Fell ideal with separable irreps to a countable π -dense family of trivial continuous-trace patches.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Lisa Orloff Clark, Astrid an Huef, and Iain Raeburn, *The equivalence relations of local homeomorphisms and Fell algebras*, arXiv:1207.2540, 2012.
- [3] J. Dixmier, *C^* -Algebras*, North-Holland, 1977.
- [4] I. Raeburn and D. P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.

- [5] Run `fa_banach_001`, lane 19, *Atomic Multiplier Charts and BCP for Fell Algebras with Separable Irreps*, solution packet `2311.14257_fell_separable_irrep_bcp`, 2026.
- [6] Run `fa_banach_001`, lane 19, *A Primitive-Space π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.